



PUSL3190 Computing Project

Project Interim Report

SPAREHUBLK (AI auto-parts market)

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Chapter 01. Introduction

1.1 Introduction

The vehicle spare parts market in Sri Lanka is growing rapidly. Many vehicle owners and mechanics depend on online platforms to find spare parts. Currently, general marketplace websites are used for buying and selling spare parts. However, these platforms are not designed specifically for vehicle components. Because of this, users face difficulties when searching for exact parts.

This project proposes an AI based online marketplace specially designed for vehicle spare parts. The system will allow users to search using text and also upload images of spare parts. The system will analyze the image using a machine learning model and suggest matching products. In addition, a price prediction model will help users understand the estimated market value of a part.

The system will be developed using the MERN stack which includes MongoDB, Express, React, and Node.js. Artificial Intelligence models will be integrated using TensorFlow.

The main goal of this project is to create a smart, reliable, and user friendly platform for spare part trading in Sri Lanka.

1.2 Problem Definition

Currently users face multiple problems when searching for spare parts online.

No.	Problem
1	Difficult to identify correct spare part
2	No image based intelligent search
3	No automated price guidance
4	Many irrelevant search results
5	Lack of proper filtering system
6	Time consuming manual searching

Table 1. problems table

- Because of these problems, buyers may purchase wrong items or pay unfair prices. Sellers also struggle to set correct prices.

1.3 Project Objectives

Main Objective

- To develop an AI powered online marketplace for vehicle spare parts in Sri Lanka.

Specific Objectives

- Develop secure user registration and login system
- Allow sellers to list spare parts with images
- Implement AI image classification model
- Develop price prediction module
- Enable real time chat between users
- Ensure system security and performance

Chapter 02. System Analysis

2.1 Facts Gathering Techniques

- Information was collected using interviews with mechanics and vehicle owners.
- Online research was conducted to analyze existing systems.
- Researched internationally and locally to check whether this type of website is available or not with these exact features.
- User search behavior was observed to understand difficulties.
- Feedback helped identify key system requirements.

2.2 Existing System

Currently users rely on general marketplace platforms. These systems allow posting and searching advertisements but lack intelligent features.

Feature	Existing System	Proposed System
Spare part listing	YES	YES
Basic keyword search	YES	YES
Chassis number decoding	NO	YES
AI image recognition	NO	YES
Price guidance	NO	YES
Smart filtering	Limited	YES(advanced)

Table 2. Comparison of Existing System and Proposed System

2.3 Drawbacks of Existing System

Existing systems are not specialized. They do not verify compatibility of parts. They lack AI based assistance. Users must depend only on keyword matching. This creates inefficiency and confusion.

Chapter 03 Requirements Specification

3.1 Functional Requirements

The system must allow users to create accounts and log in securely. Sellers must be able to add, edit and delete spare part listings. Buyers must be able to search parts using keywords and upload images. The system must analyze images and return matching results. The system must predict estimated price using machine learning. Users must be able to communicate using real time messaging. The admin must manage users and listings.

Module	Description
Listing Management	Add and manage spare parts
Search Module	Text and image based search
Price Prediction	Estimate market price
Chat System	Real time communication
User Management	Registration and authentication
Admin Panel	System management

Table 3. Key Functional Modules

3.2 Non Functional Requirements

The system must provide fast response time under three seconds. It must ensure secure authentication using encrypted passwords. The interface must be simple and user friendly. The system must be reliable and scalable.

3.3 Hardware and Software Requirements

Component	Requirement
Processor	Intel i5 or above
RAM	Minimum 8GB
Backend	Node.js with Express
Frontend	React Vite
Database	Mongo DB
AI Framework	TensorFlow

Table 4. Development Environment

3.4 Networking Requirements

The Spare Hub Platform is a web based system. Because of that, networking plays an important role in the system. The application needs proper internet connection and server communication to work smoothly.

The system follows a client server architecture. Users access the system using a web browser. The frontend communicates with the backend server through API requests. The backend then connects to the MongoDB database to store and retrieve data.

Below are the networking requirements of the system.

Internet Access

- Users must have a stable internet connection
- Minimum broadband connection is required
- Image uploading requires good upload speed
- Slow internet may affect system performance

Server Requirements

- Backend server must be hosted online
- Server must have public IP address or domain
- Server should run continuously without downtime
- Proper firewall configuration is required

Communication Protocol

- System uses HTTP and HTTPS protocols
- HTTPS is recommended for security
- Data between frontend and backend is transferred in JSON format

API Communication

- Frontend sends REST API requests to backend
- Backend processes request and sends response
- Authentication is handled using JWT
- Passwords are encrypted using Bcrypt

Database Connection

- Backend connects securely to MongoDB
- Database connection string must be protected
- Remote database access must be restricted
- Database should allow secure authentication

Security Requirements

- HTTPS encryption
- JWT based authentication
- Password hashing
- Firewall protection
- Restricted database access

The network flow of the system is simple and organized. The user browser connects to the frontend application. The frontend communicates with the backend server. The backend handles business logic and connects to the MongoDB database.

Chapter 04 Feasibility Study

4.1 Operational Feasibility

The proposed system is easy to operate because users are already familiar with online marketplaces. The interface will be designed clearly to support both buyers and sellers.

4.2 Economical Feasibility

The system uses open source technologies. Therefore software cost is minimal. The only costs include hosting and domain registration. Since this is an academic project, development cost is not significant.

4.3 Technical Feasibility

MERN stack is widely used for modern web applications. TensorFlow supports image classification and regression models. Technical resources and documentation are available. Therefore the project is technically feasible.

Chapter 05 System Architecture

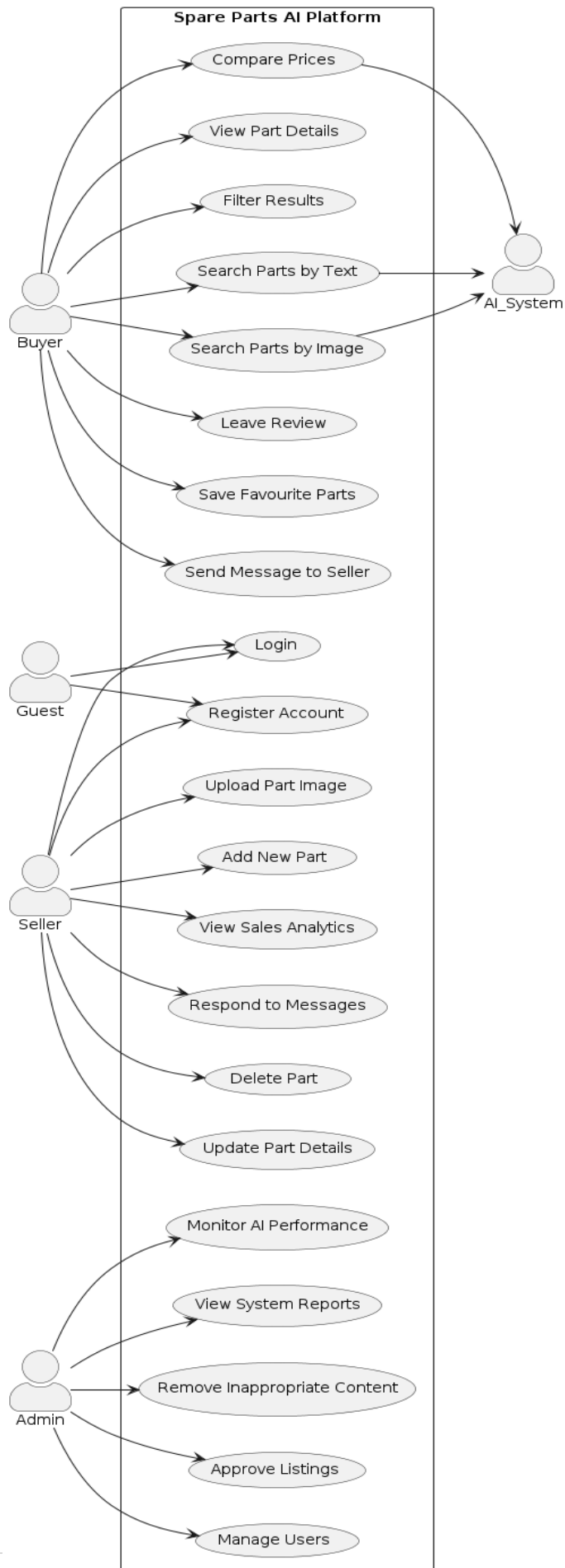
5.1 Use Case Diagram

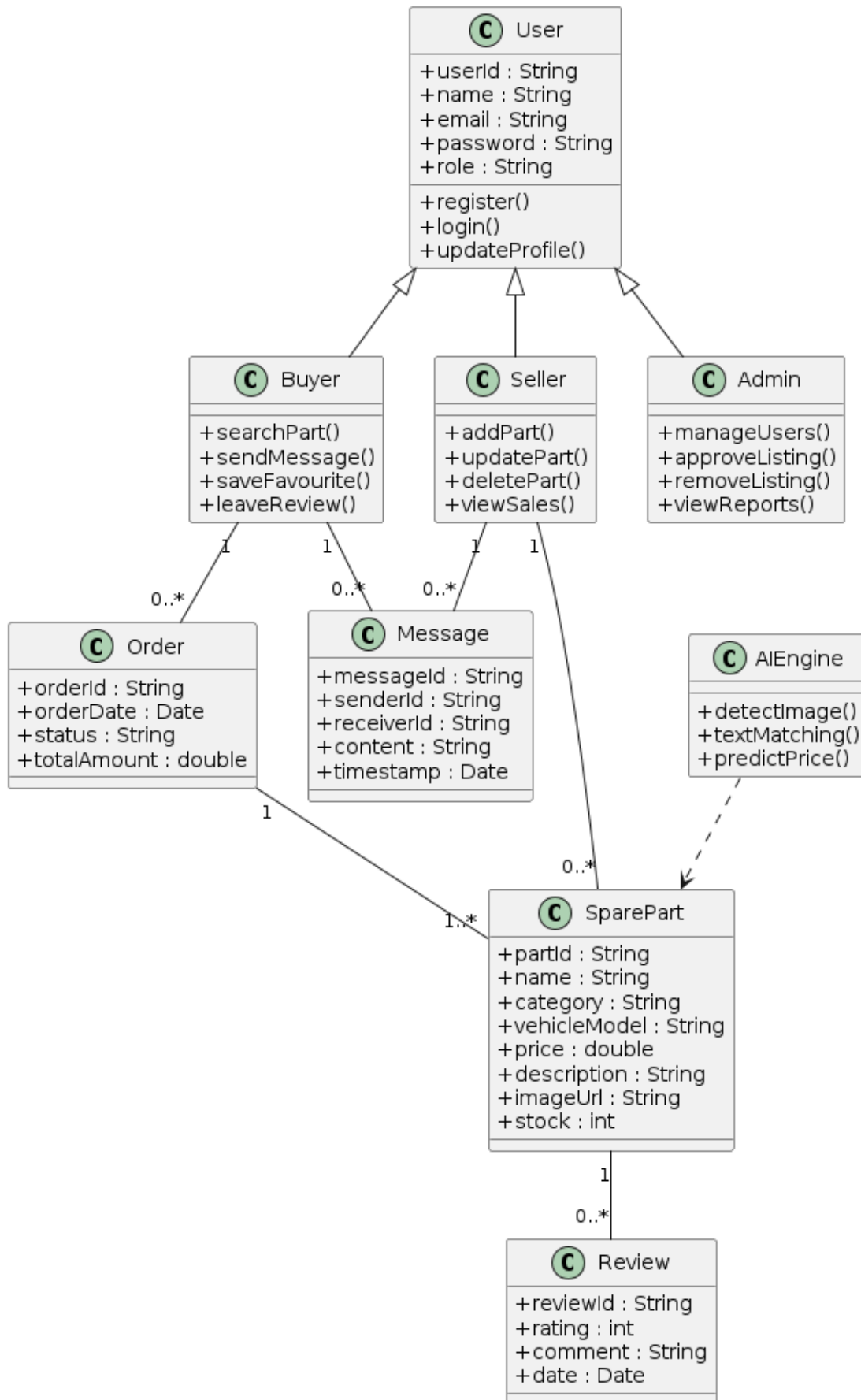
5.2 Class Diagram of Proposed System

5.3 ER Diagram

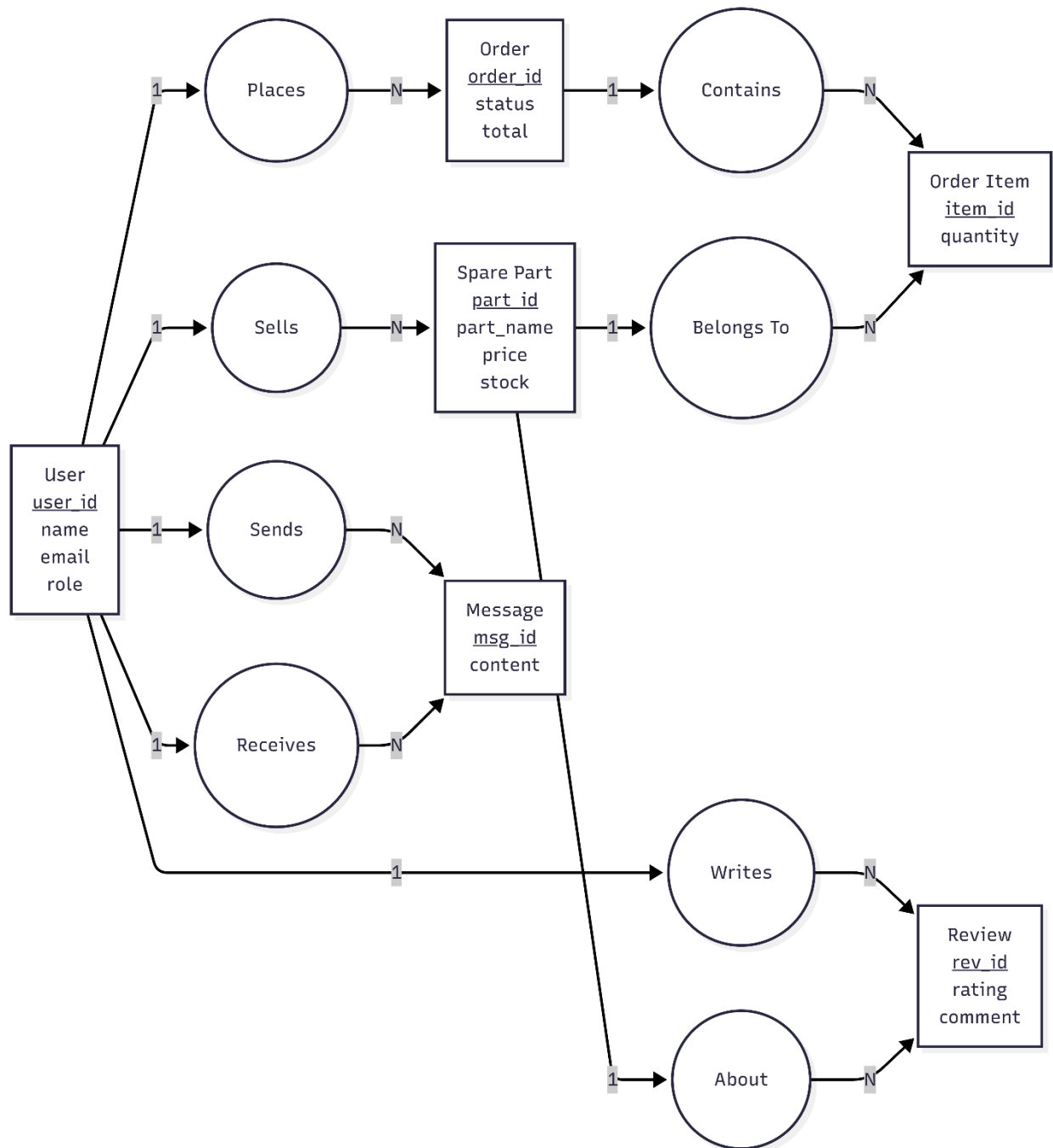
5.4 High-level Architectural Diagram

5.5 Networking Diagram

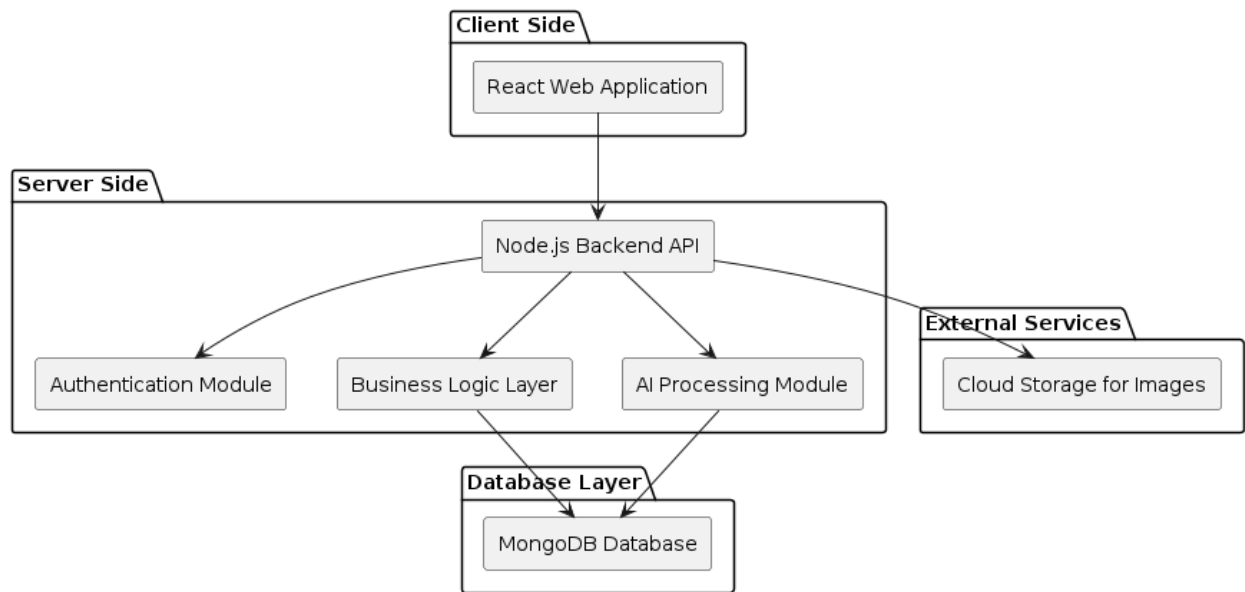




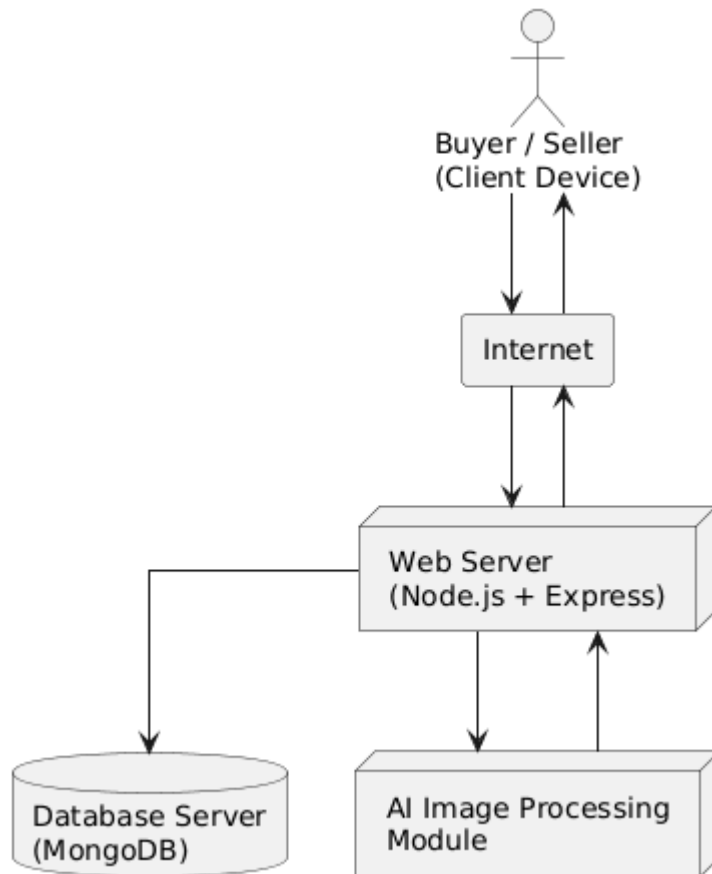
5.2 Class Diagram of Proposed System



5.3 ER diagram



5.4 High-level Architectural Diagram



5.5 Networking Diagram

Chapter 06 Development Tools and Technologies

6.1 Development Methodology

The development of this system follows the Agile methodology. Agile is selected because it allows the project to be developed step by step instead of building everything at once.

The system is divided into smaller parts. Each part is developed, tested, and improved before moving to the next part. This makes it easier to identify mistakes early and fix them without affecting the whole system.

The main reasons for choosing Agile are:

- It supports gradual improvement
- It allows feedback during development
- It reduces risk
- It helps manage time effectively

During development, features such as login, spare part listing, and search were completed first. Advanced features such as image recognition and price prediction are being added gradually.

This method ensures that the system remains stable while new functions are added.

6.2 Programming Languages and Tools

The system is developed as a web based platform. Different tools are used for different purposes.

Component	Technology Used	Purpose
Frontend	React Vite with Tailwind CSS	To create user interface
Backend	Node.js	To handle server operations
Backend Framework	Express	To manage system routes
Database	MongoDB	To store system data
Version Control	Git	To track project changes
Development Tool	Visual Studio Code	To write and manage code

Table 6. Technologies Used in the System

6.3 Third Party Components and Libraries

To improve system quality and security, some external libraries are used.

- Authentication
 - ✓ JSON Web Token is used to secure user login. It ensures that only authorized users can access protected pages.
- Password Security
 - ✓ Passwords are stored in encrypted form using a hashing library. This improves system security and protects user data.
- Image Recognition
 - ✓ TensorFlow is used to identify spare parts from uploaded images. It compares image patterns with stored data to suggest matching spare parts.

6.4 Algorithms

The system includes important logical processes that support intelligent features.

Image Identification Process

- When a user uploads an image, the system analyzes the image and compares it with trained data. Based on similarity, it suggests the most relevant spare part.

Text Search Process

- When a user types a keyword, the system searches the database and displays results that match the entered text.

Price Suggestion Process

- The system studies previous prices of similar spare parts and suggests an estimated price range. This helps buyers understand whether the listed price is reasonable.

These processes improve user experience and make the system smarter than a normal buying and selling platform.

Chapter 07 Implementation Progress

This chapter explains the current progress of the system. It shows that actual development has already started and important features are completed.

7.1 Development Environment Setup

The development environment was prepared successfully.

The following steps were completed:

- Installation of Node.js
- Installation of MongoDB
- Creation of project folder structure
- Database connection testing
- API testing using development tools

The frontend and backend are separated into different folders to maintain proper organization.

7.2 Implemented Features

The following core features have been successfully implemented.

Feature	Status
User Login / Sign Up	Complete
Core shop	Complete
Sell spare parts	Complete
User Dashboard	Complete
Search parts	Complete
Basic UI	Complete

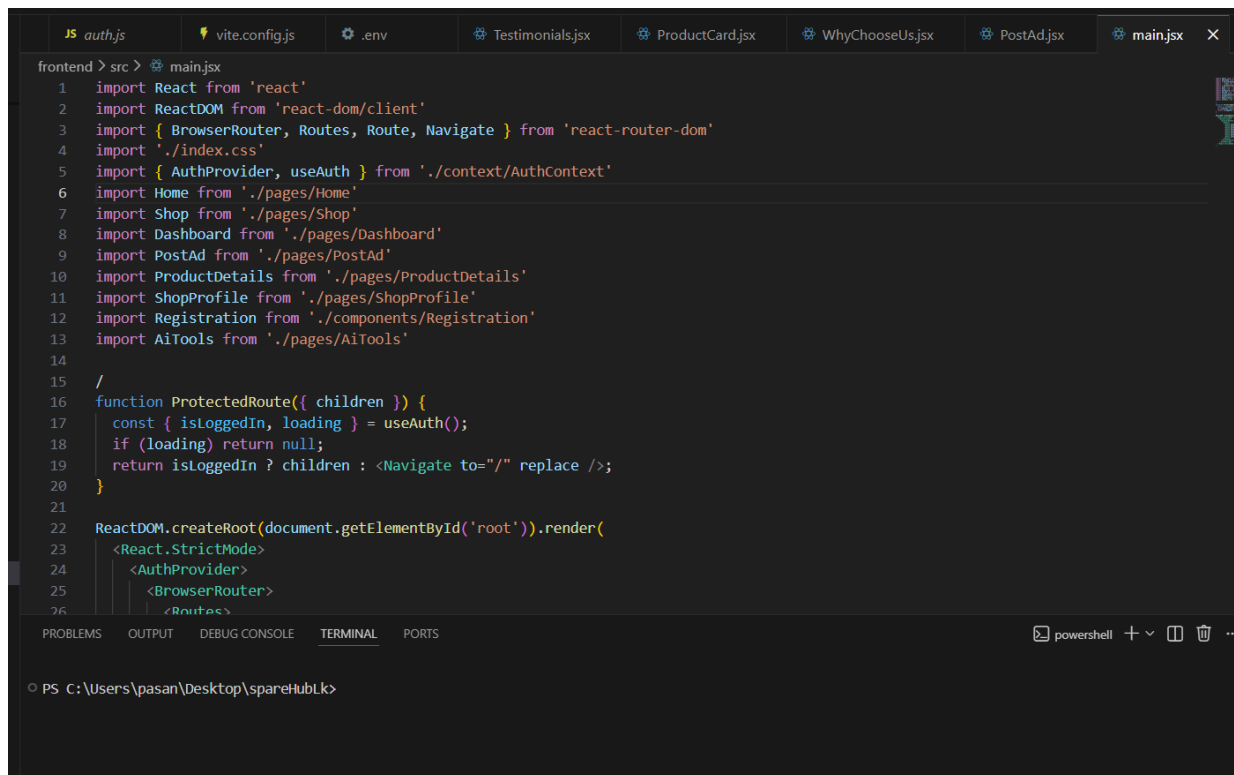
Table 7 Completed Features

Currently, users can:

- Register and log in
- Add spare parts
- View spare parts
- Search spare parts
- Edit and remove spare parts

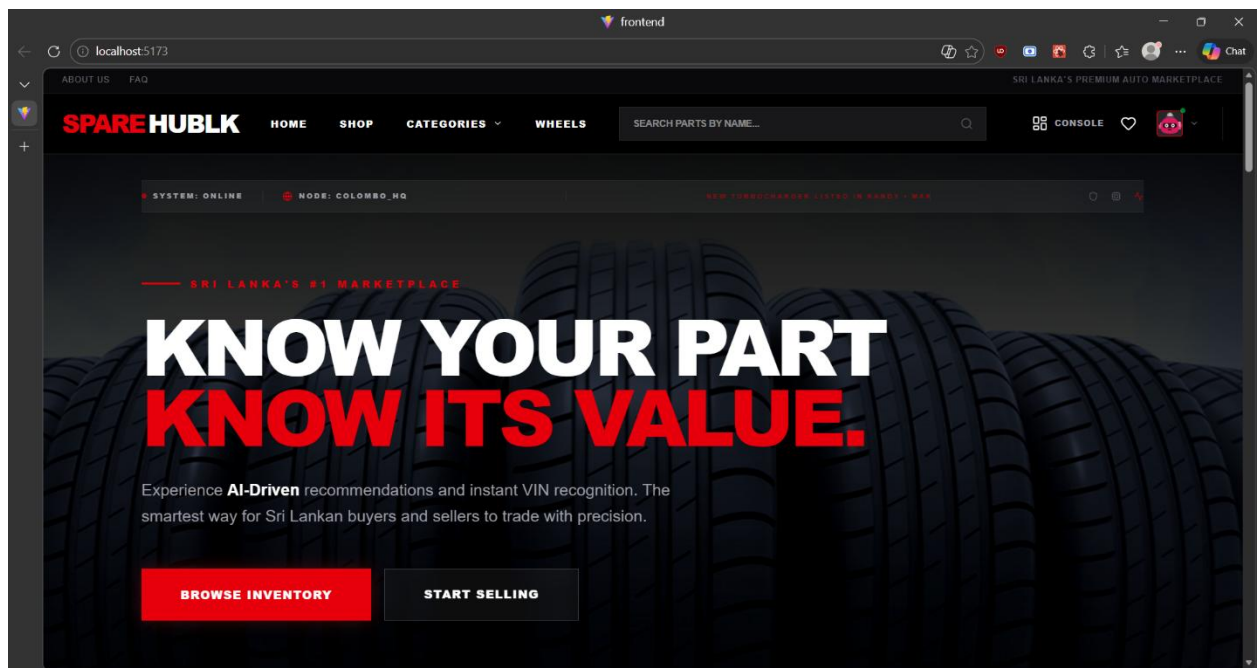
7.3 Screenshots

Main page connections of SPAREHUBLK

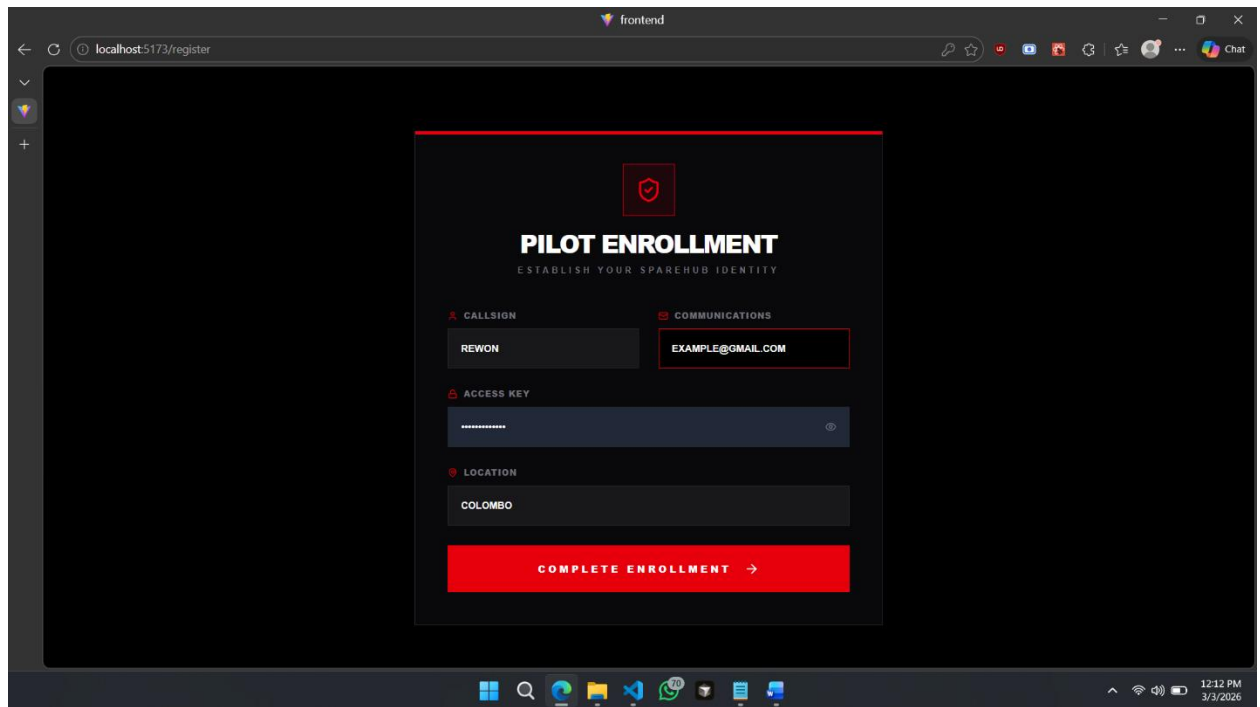


```
1 import React from 'react'
2 import ReactDOM from 'react-dom/client'
3 import { BrowserRouter, Routes, Route, Navigate } from 'react-router-dom'
4 import './index.css'
5 import { AuthProvider, useAuth } from './context/AuthContext'
6 import Home from './pages/Home'
7 import Shop from './pages/Shop'
8 import Dashboard from './pages/Dashboard'
9 import PostAd from './pages/PostAd'
10 import ProductDetails from './pages/ProductDetails'
11 import ShopProfile from './pages/ShopProfile'
12 import Registration from './components/Registration'
13 import AiTools from './pages/AiTools'
14
15 /
16 function ProtectedRoute({ children }) {
17   const { isLoggedIn, loading } = useAuth();
18   if (loading) return null;
19   return isLoggedIn ? children : <Navigate to="/" replace />;
20 }
21
22 ReactDOM.createRoot(document.getElementById('root')).render(
23   <React.StrictMode>
24     <AuthProvider>
25       <BrowserRouter>
26         <Routes>
```

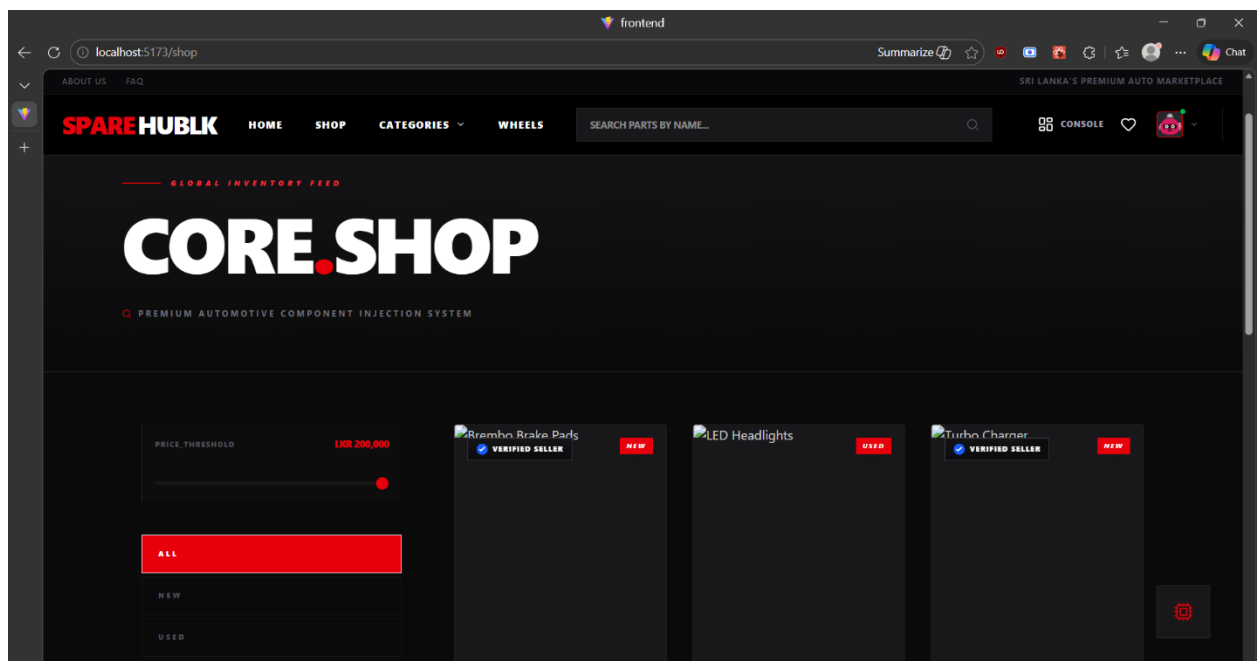
Hero page of SPAREHUBLK



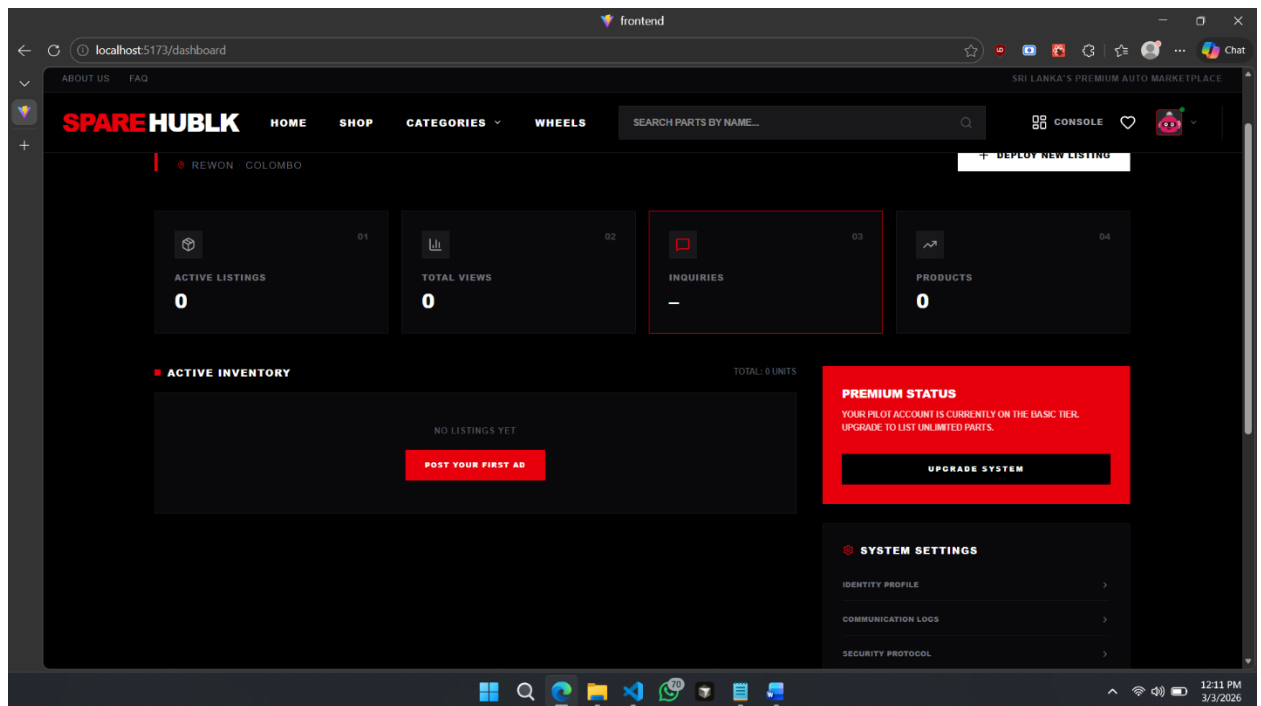
User Login / User registration



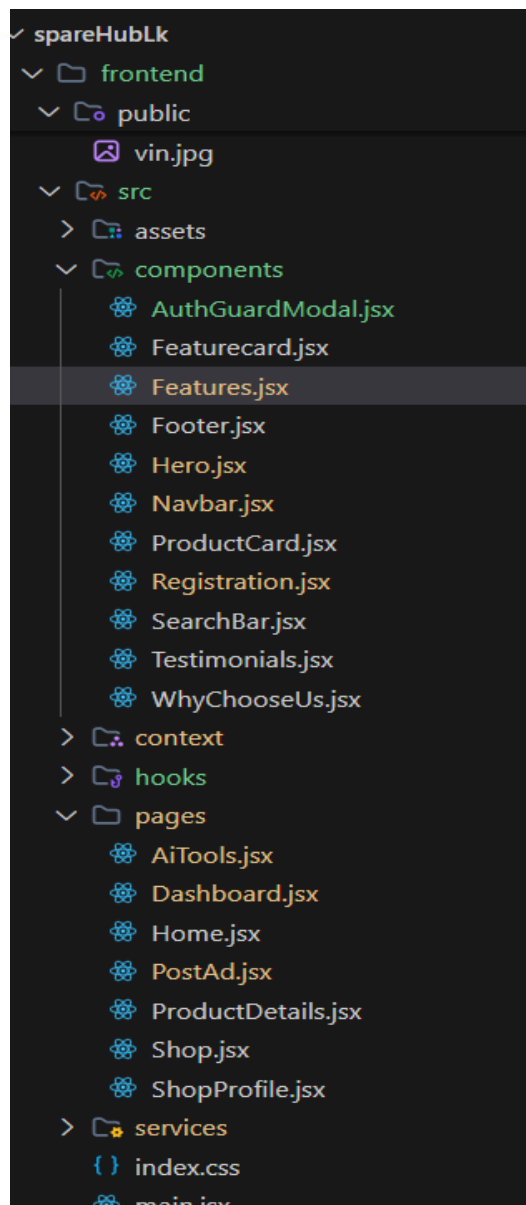
Shop page of SPAREHUB



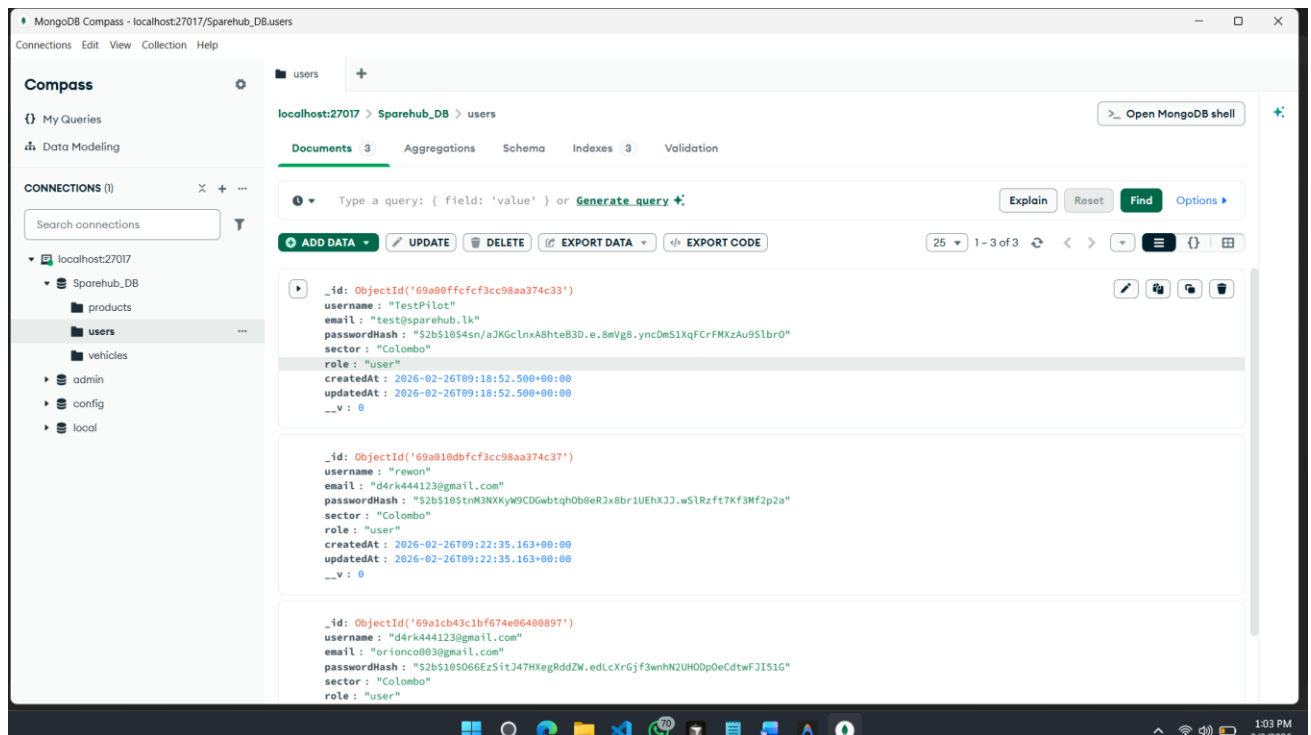
Seller Dashboard



File organization



Test user data on database



7.4 Challenges Encountered and Solutions

During the development of the Spare Parts AI Platform, several practical challenges were encountered. These challenges helped improve technical understanding and problem solving skills. The major issues and their solutions are explained below.

1. Database Connection Failure

Challenge - At the beginning of backend development, the system failed to connect to the MongoDB database. The server was running, but data could not be stored or retrieved.

Cause - The database connection string was incorrectly configured, and environment variables were not properly loaded.

Solution - The connection string was carefully checked and corrected. Environment configuration files were reviewed and updated. After restarting the server and verifying the database port, the connection was successfully established.

2. Cross-Origin Request Errors

Challenge - When the frontend tried to send requests to the backend, the browser blocked the request and displayed an error.

Cause - The frontend and backend were running on different ports, which caused cross-origin request restrictions.

Solution - Cross-origin request handling was enabled in the backend configuration. After allowing requests from the frontend port, the communication between frontend and backend worked correctly.

3. User Authentication Issues

Challenge - After login, users were not able to access protected pages. The system kept redirecting them back to the login page.

Cause - The authentication token was not properly stored or verified.

Solution - The login logic was reviewed. The token was correctly generated and stored in the browser. Token verification was properly added to protected routes. After fixing this, role based access worked correctly.

4. Password Encryption Errors

Challenge - Initially, passwords were saved in plain text format in the database, which is insecure.

Cause - The password hashing process was not implemented before saving user data.

Solution - Password hashing was implemented before storing user details. During login, the entered password is now compared with the encrypted password stored in the database. This improved system security.

The challenges faced during development were technical, structural, and time related. Each problem was carefully analyzed and solved through testing, debugging, and restructuring. These experiences improved development skills and strengthened understanding of full-stack system integration.

7.5 Current System Limitations

Although the core functionalities of the Spare Parts AI Platform are fully operational, several limitations still exist at the current development stage. These limitations are identified to ensure transparency and continuous improvement.

- One key limitation is the accuracy of the image recognition feature and vehicle Chassis number decoder. While the system can identify spare parts based on uploaded images, the prediction accuracy depends heavily on the

quality and variety of the training data and for the chassis decoding feature two API s are used to fetch data.

- The real-time chat feature between buyers and sellers is partially implemented. Basic message feature is available, but it's functionality is under development.
- In addition, the user interface, although functional and responsive, can be further improved in terms of visual refinement and user experience design. Minor alignment and usability enhancements are planned for future updates.

These limitations are part of the normal development process and will be addressed in the next phase to enhance overall system quality and performance.

Chapter 8 Discussion

Summary of the Report

This interim report explains the design and current progress of the Spare Parts AI Platform. The main goal of this system is to create an easy and smart platform where users can buy and sell spare parts online.

In this report, the problem was clearly explained and the weaknesses of the current manual or traditional systems were discussed. After that, system requirements were identified to make sure the new system meets user needs. A feasibility study was also done to check whether the project is possible in terms of time, cost, and technical knowledge.

The system architecture was designed to separate the frontend, backend, and database properly. This makes the system more organized and easier to manage. The tools and technologies used for development were also explained in detail.

From the implementation side, the main features such as user registration, login, adding spare parts, editing items, and searching have already been completed and tested. These features show that the core system is working properly.

Advanced features like image recognition and price suggestion are also added to the system. However, they still need improvement to increase their accuracy and performance.

Overall, the project is progressing according to the plan. The system is working at a functional level, and it proves that the project idea is practical and achievable within the given time period.

What Has Changed from the Proposal

Compared to the original proposal, several refinements were made during development.

Initially, the project aimed to implement all artificial intelligence features at an advanced level from the beginning. However, during implementation, the development priority was adjusted to focus first on core selling functionalities to ensure system stability.

The image recognition Chassis decoder and price prediction features were simplified in the interim stage to allow gradual improvement instead of full scale complex implementation.

Additionally, the system structure was reorganized to improve maintainability and scalability.

These changes were necessary to ensure realistic development within the given timeframe while maintaining system quality.

Future Plans / Upcoming Work

The next development phase will focus on improving performance, accuracy, and user experience.

Planned improvements include:

Future Plans / Upcoming Work

The next development phase will focus on improving performance, accuracy, and user experience.

Planned improvements include

- Increasing image recognition accuracy by expanding training data
- Enhancing the price prediction model with more historical data
- Completing real-time chat functionality with notification support
- Improving user interface design and user experience
- Conducting full system testing and performance optimization
- Deploying the system on a cloud hosting platform
- Using 2 or more API s to fetch Srilanka wide chassis number decoder

References

MongoDB documentation
Tailwind CSS documentation
W3school
www.nhtsa.gov
ikman.lk

riyasewana.lk

Online technical guides and tutorials (AI model training)

Appendixes

- Vehicle chassis number API (only USA)
Since spare hub is a srilankan based website , had to use more API is to fetch data ,
but still NHTSA API works

